ABARES publishes working papers to provide its research teams with the opportunity for external peer review of ongoing research projects.

These papers are technical in nature and written in the style of a journal paper. Some of these working papers may be submitted to and accepted by a journal, or they may continue development as part of a research report intended for formal publication as an ABARES product.

This working paper develops a method to predict monthly irrigation water demands in the Murray-Darling Basin. This work is a key input for simulation analysis in hydro-economic modelling. In the future this bio-economic approach could provide a foundation for integrated hydro-economic models to help analyse complex water policy issues, including environmental water management, water market design and climate change adaptation.

Published 22 December 2025

This working paper demonstrates the feasibility and utility of developing integrated monthly Hydro-Economic models of the Murray-Darling Basin. To demonstrate the capability of these models they have been applied to a range of future climate change scenarios to project possible implications for water markets for the southern Murray-Darling Basin. This work has to potential to assist water planning in the future.

Published 22 December 2025

This report presents the underlying measurement framework used for the ABARES Farmland Price Indicator. It outlines the use of administrative data supplied by CoreLogic in a stratification model to estimate average Australian farmland prices. It replaces an earlier version that is now obsolete due to recent methodological updates.

Published 28 November 2024

This paper is the first to propose and apply a method for the estimation of Australian vegetable productivity and provides a foundation for future efforts to estimate productivity for Australian horticulture.

Published 30 April 2024

This working paper presents methods to improve analysis and policy advice relating to the impacts of climate policies in computable general equilibrium models, with relevance to agriculture, forestry and land use. The improvements include better representation of livestock herds and the inclusion of resource costs associated with the abatement and land-based sequestration activities, making the model more representative of real-world economic relationships.

Australia imposes regulations on goods that arrive in the country and have the potential to introduce exotic pests and diseases.

In this paper, we extend that foundational work to include uncertainty and inspection sensitivity, both of which are highly relevant to Australian border operations where samples from arriving consignments, with low but variable levels of contamination, are selected and inspected.

We provide analytical distributions and statistics for all processes of the CSP-1 cycle, including expressions for variance, for the probability that leakage occurs, and for the volume of leakage conditional on occurrence. The effect of inspection sensitivity is included and explicit.

Australia imposes regulations on goods that arrive in the country and have the potential to introduce exotic pests and diseases.

Continuous sampling plans (CSP) are operational biosecurity systems commonly implemented at the border to manage risk and, simultaneously, keep regulatory inspection costs low.

Currently, CSP-1 systems are underpinned by the results and proposed design-criteria in the classical work of Dodge (1943), and CSP-2 and CSP-3 systems are underpinned by that in Dodge and Torrey (1951).

This paper provides fast, accurate and accessible solutions to replace the need for simulations in many applications the continuous sampling plan 2 and 3 systems used in the management of Australia's biosecurity system. This will aid in enhancing system management decisions.

There is considerable debate around discount rates and the treatment of risk in economic analysis. Since the 1980s, most government economic appraisal guidelines in Australia have adopted a consistent approach set out in government economic appraisal guidelines. However, support for this position has eroded over time.

Critics have argued that the discount rate should be updated to reflect changing economic conditions and suggested, more fundamentally, that the theoretical basis of the standard approach is flawed. If these criticisms hold, we are likely to be underestimating the merits of long-term projects versus short term projects and the merits of projects that reduce risk versus projects that increase risk, with implications for the quality of investment decisions. This report explores both sides of the debate to provide rigorous and practical guidance on how economists should approach discount rates and the treatment of risk in their analysis of agricultural projects.

Published 1 April 2022

As tariffs in Australia's major trading partners have fallen, trade negotiators are increasingly focussed on the non-tariff elements of trade agreements. A negotiator is faced with the policy challenge of determining what non-tariff measures (NTMs) and products to focus on in bilateral and multilateral negotiations. This means that qualitative and quantitative advice on the trade effects of different measures can inform one aspect of a negotiator’s multifaceted decision-making process to prioritise and allocate scarce negotiating resources. This paper outlines a defensible application of theory that leads to a ‘proof of concept’ quantification methodology for product-level bilateral trade effects of NTMs.

Published 23 February 2022

Recent shifts in the Australian climate including both higher temperatures and lower winter rainfall, have had significant effects on the agriculture sector. Despite these recent trends, there remains uncertainty over the future climate and its potential impacts on Australian farm businesses. In this study, a statistical model of Australian cropping and livestock farms is applied to simulate the potential effects of climate change on farm profits.

This farm model is combined with a range of downscaled projections for temperature and rainfall by 2050. The results provide an indication of adaptation pressure: showing which regions, sectors and farm types may be under greater pressure to adapt or adjust to climate change. The future climate scenarios produce a wide range of outcomes, with simulated changes in average farm profits (without adaptation) ranging from -2% to -32% under an 'intermediate' global emissions scenario (RCP4.5) and -11% to -50% under a 'high' global emissions scenario (RCP8.5), relative to the reference period climate of 1950 to 2000.

Published 29 July 2021.

This paper presents an economic model of water markets and irrigated agriculture within the Australian southern Murray-Darling Basin (sMDB). A unique data set, detailing irrigation activity, water supply and prices over the period 2000-01 to 2018-19, is used to estimate a set of water demand functions by catchment region and irrigation activity. The resulting model can realistically simulate the market prices of water, carryover volumes and trade flows across the sMDB along with irrigation water use and area planted by region and activity.

The model is applied to simulate distributions for water prices in the sMDB, and to estimate the economic benefits of interregional water trading and carryover. Water markets are estimated to generate benefits to water users in the sMDB of $117m per year on average (around 12 per cent of the market value of water rights in the region). The benefits of water markets are reduced by around half in the absence of carryover.

The Defining drought from the perspective of Australian farmers working paper presents a new drought indicator as an alternative to traditional rainfall measures.

The new indicator is found to more accurately reflect the effects of drought on farm profits than simple rainfall measures, accounting for the unique circumstances of each farm and the effects of movements in domestic commodity prices (i.e. high grain and hay prices).

The Defining drought risk report also presents data on farmer self-assessments of drought. These data show differences in farmer perception of drought once variation in climate conditions has been controlled for. This includes a tendency for farmers in New South Wales and Queensland to be more likely to self-assess as 'in-drought' than farmers in Western Australia. There is also a trend for current farmers to be less likely to assess as in-drought than farmers in the past.

ABARES has developed a new model, farmpredict which can simulate the effects of price and climate variability on the production and profitability of Australian broadacre farms. farmpredict simulates the production of outputs, the use of inputs and changes in farm stocks at a farm level, given information on farm fixed inputs, input and output prices and prevailing climate conditions.

This paper presents a new farm level microsimulation model of irrigation activity within the southern Murray-Darling Basin (sMDB), developed by ABARES on behalf of the Department of Agriculture, Fisheries and Forestry.

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